with convolutional neural networks (R-CNN) for multiorgan segmentation. Manual evaluation of calcium was carried out using proprietary software. This study used 80 patients to train the segmentation model and randomly selected 1442 patients were used for the validation of the algorithm.

Results Our method correctly classifies the risk group in 78.8% of the cases and classifies the subjects in the same group. F-score is measured as 0.78; 0.71; 0.81; 0.82; 0.92 in calcium score categories 0(CS:0), I (1–99), II (100–400), III (400–1000), IV (>1000), respectively. 79% of the predictive scores lie in the same categories, 20% of the predictive scores are one category up or down, and only 1.2% patients were more than one category off. For the presence/absence of coronary artery calcifications, our deep learning model achieved a sensitivity of 90% and a specificity of 94%.

Conclusion Fully automated model shows good correlation compared with reference standards. Automating the process could improve diagnostic ability, risk categorization, facilitate primary prevention intervention, improve morbidity and mortality, and decrease healthcare costs.

Conflict of Interest Advisor to KARDIOLABS.AI, who developed the AI Algorithm

179 DERIVING MEAN RIGHT ATRIAL PRESSURE FROM CMR: DEVELOPMENT OF A MODEL FROM THE ASPIRE REGISTRY

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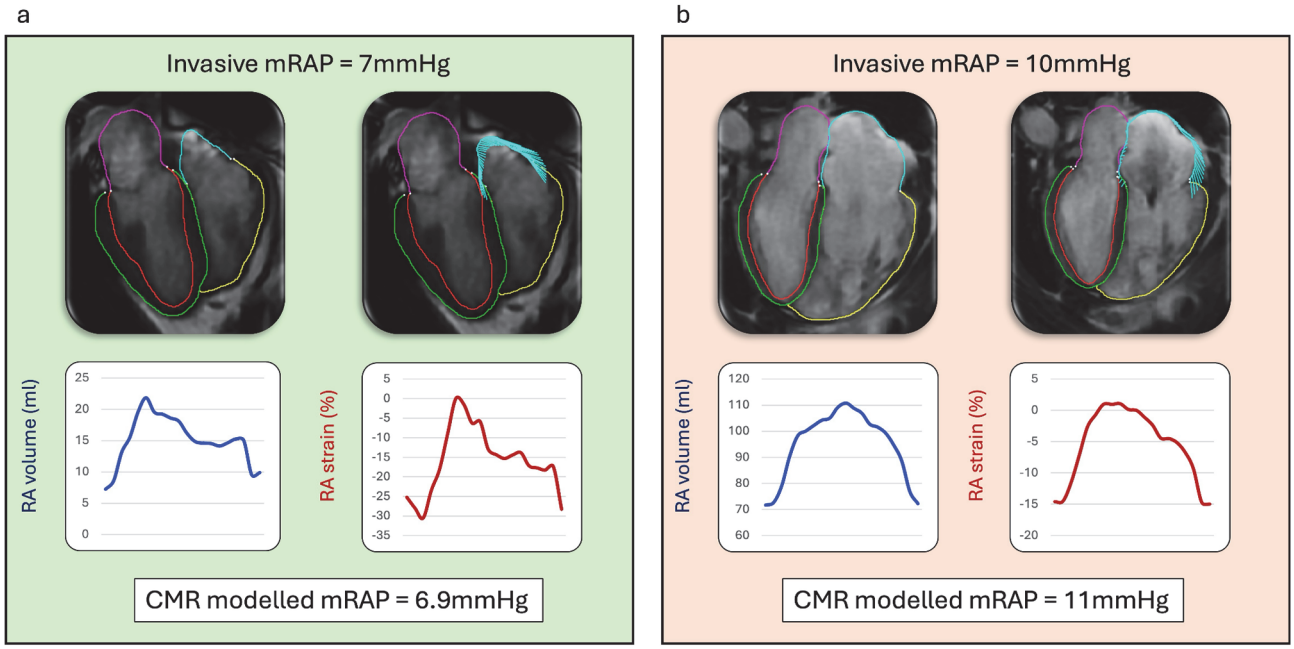
Background Right atrial pressure (RAP) corresponds to fluid status and preload and is also important in prognostication for patients with heart failure and pulmonary hypertension. RAP can be measured invasively or non-invasively, but cannot currently be estimated by cardiac MRI (CMR). This study used paired CMR and invasive right heart catheter (RHC) measurements to develop a model to predict RAP from CMR.

Methods The ASPIRE registry consists of patients referred for assessment of dyspnoea to Sheffield Teaching Hospitals between 2012 and 2020. Inclusion criteria were age >18 years, signs and symptoms of heart failure and adequate CMR image quality. Patients diagnosed with pulmonary arterial hypertension were excluded. RHC and CMR were performed in the same 24 hour period.

CMR was performed with a 1.5 T GE HDx scanner. Chamber dimensions, strain, ejection fraction and stroke volume were acquired from 2- and 4- chamber cine-images using operator reviewed AI contours in MASS research software (Fig 1).

Abstract 179 Table 1 Study demographics of the cohort (n=672). HFpEF = Heart Failure with preserved ejection fraction, HFmrEF = Heart Failure with moderately reduced ejection fraction, HFrEF = Heart Failure with reduced ejection fraction

	mRAP ≤ 8 mmHg	mRAP > 8 mmHg	P-value
N (%)	295 (44%)	377 (56%)	
Age (years)	64±14	68±12	<0.01
Male sex (%)	109 (37%)	174 (46%)	0.02
HFpEF (%)	150 (51%)	195 (52%)	0.82
HFmrEF (%)	7 (2.4%)	18 (5%)	0.10
HFrEF (%)	5 (1.7%)	10 (3%)	0.40
Heart rate (bpm)	72±15	70±16	0.05
Systolic blood pressure (mmHg)	142±25	143±27	0.52
Diastolic blood pressure (mmHg)	76±11	79±13	0.01



Abstract 179 Figure 1 Two case examples. **Panel a** | Normal invasive right atrial pressure (7 mmHg). **Panel b** | A patient with a dilated right atrium and significantly compromised right atrial function (volume curves are flattened) with raised invasive mRAP (10 mmHg)

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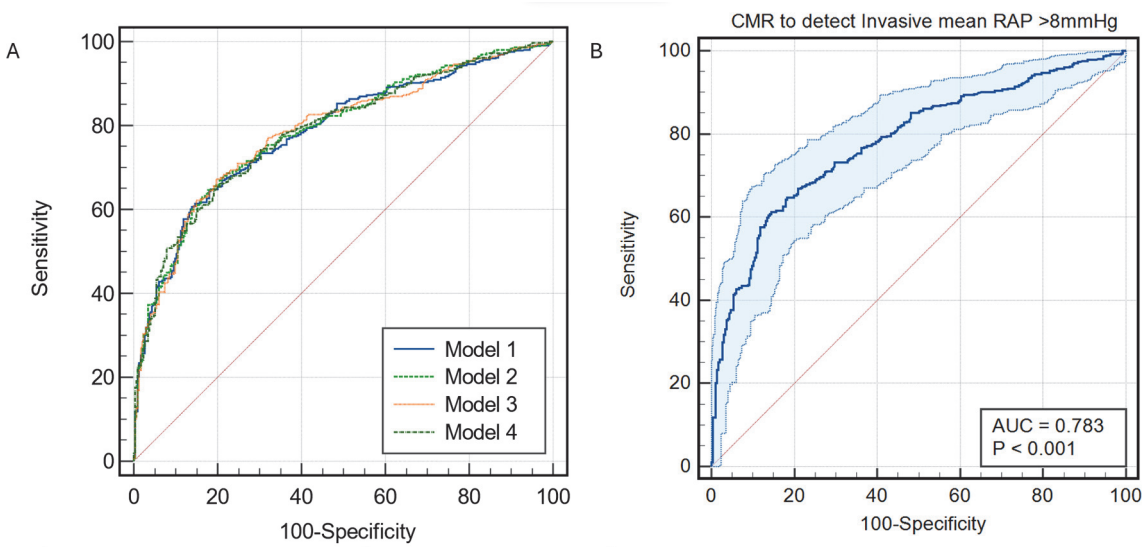

	AUC	95% CI
Model 1	0.783	0.749 to 0.813
Model 2	0.785	0.752 to 0.815
Model 3	0.786	0.753 to 0.817
Model 4	0.786	0.753 to 0.816

Pairwise comparison of all ROC curves using DeLong method:

P = non-significant

Abstract 179 Figure 2 Multivariable models. **Panel A** | Four physiological models to estimate invasive mRAP. Model 1 only incorporates RA end systolic volume, Model 2 incorporates RA End systolic volume and peak strain, whereas Model 3 incorporates RA end systolic volume and RV volumes. Model 4 includes RA end systolic volume, body mass index and sex. There is no significant difference in diagnostic power using any model over Model 1. **Panel B** | Model 1 ROC with 95% confidence interval presented in light blue with an mRAP threshold of 8 mmHg

Abstract 179 Table 2. Pearson correlation of several CMR indices with invasive mean right atrial pressure (mmHg) RA = Right Atrial, RV = Right Ventricular

	Correlation coefficient	P-value
RA End Diastolic Volume (ml)	0.553	<0.01
RA End Systolic Volume (ml)	0.579	<0.01
RA Ejection Fraction (%)	-0.545	<0.01
RA Stroke Volume (ml)	0.085	0.03
RA peak strain (%)	0.511	<0.01
RV End Diastolic Volume (ml)	0.442	<0.01
RV End Systolic Volume (ml)	0.416	<0.01
RV Stroke Volume (ml)	0.257	<0.01
RV Ejection Fraction (%)	-0.223	<0.01

Associations between invasive mean RAP (mRAP) and CMR metrics were assessed with Pearson's product-moment correlation coefficient. Stepwise multiple linear regression was used to develop models to predict mRAP. These were compared with receiver-operator curve analysis and DeLong's test.

Results The cohort was made up of 672 patients, divided based upon invasive mRAP ≤ 8 mmHg (44%) and mRAP > 8 mmHg (56%) (table 1). Those with higher mRAP tended to be older, more likely to be male and have a higher diastolic blood pressure but there was no difference in rates of different types of heart failure (table 1).

Metrics with the strongest correlation to invasive mRAP were right atrial (RA) dimensions, strain and ejection fraction with moderate correlation to right ventricular (RV) measurements (table 2). Four multivariable models were developed. Model 1 contained RA dimensions. RA strain was added to this to create Model 2. Model 3 contained RA and RV dimensions whilst model 4 incorporated RA end systolic volume (RAESV), the single strongest CMR predictor, and adjusted it for body mass index and sex.

All four models had similar predictive capability (Fig 2 Panel A). Model 1 identified RAESV as the only variable required for mRAP prediction (coefficient = 0.06, $p < 0.001$) and was therefore preferred for simplicity. Using a threshold of mRAP >8 mmHg ROC analysis demonstrated an area under the curve of 0.78 (95% Confidence interval 0.75 to 0.81) (Fig 2 Panel B).

Conclusions mRAP can be estimated with moderate confidence from CMR RAESV. Further studies are required to validate this model externally, including larger cohorts with reduced ejection fraction, and determine its clinical significance. Alternative clinical and MRI data could be explored to further enhance the models predictive capability. Application of this technique could enhance the role of CMR in non-invasive haemodynamic assessment and a wide range of cardiovascular pathophysiology.

Conflict of Interest None

180 MULTIPARAMETRIC CARDIOVASCULAR MAGNETIC RESONANCE IMAGING BEFORE INVASIVE CORONARY ANGIOGRAPHY IN SUSPECTED NON-ST-SEGMENT ELEVATION MYOCARDIAL INFARCTION CAN CHANGE MANAGEMENT IN OVER ONE-THIRD OF THE PATIENTS

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